In the table below we give some functions, their derivatives and their differentials side by side for comparative purpose.

S. No.	Function	Derivative	Differential
1	$f(x)=x^n$	$f'(x)=nx^{n-1}$	$df=nx^{n-1}dx$
2	$f(x)=\cos(x^2+7x)$	$f'(x)=-\sin(x^2+7x)(2x+7)$	$df=-\sin(x^2+7x)(2x+7)dx$
3	$f(x)=\cot(x^2)$	$f'(x)=-\operatorname{cosec}^2(x^2)2x$	$df=-\operatorname{cosec}^2(x^2)2x\,dx$
4	$f(x)=\sin^{-1}(x)$	$f'(x)=\frac{1}{\sqrt{1-x^2}}$	$df=\frac{1}{\sqrt{1-x^2}}dx$
5	$f(x)=\tan^{-1}x$	$f'(x)=\frac{1}{1+x^2}$	$df=\frac{1}{1+x^2}dx$
6	$f(x)=e^{x^3-5x+7}$	$f'(x)=e^{x^3-5x+7}(3x^2-5)$	$df=e^{x^3-5x+7}(3x^2-5)dx$
7	$f(x)=\log(x^2+1)$	$f'(x)=\frac{2x}{x^2+1}$	$df=\frac{2x}{x^2+1}dx$

Next we look at the properties of differentials. These results easily follow from the definition of differential and the rules for differentiation. We give a proof for (5) below and the other proofs are left as exercises.

Properties of Differentials

Here we consider real-valued functions of real variable.

(1) If f is a constant function, then $df=0$.

(2) If $f(x)=x$ identity function, then $df=1dx$.

(3) If f is differentiable and $c\in\mathbb{R}$, then $d(cf)=cf'(x)dx$.

(4) If f,g are differentiable, then $d(f+g)=df+dg=f'(x)dx+g'(x)dx$.

(5) If f,g are differentiable, then $d(fg)=fdg+gdf=(f(x)g'(x)+f'(x)g(x))dx$.

(6) If f,g are differentiable, then $d(f/g)=\frac{gdf-fdg}{g^2}=\frac{g(x)f'(x)-f(x)g'(x)}{g^2(x)}dx$, where $g(x)\neq 0$.

(7) If f,g are differentiable and $h=f\circ g$ is defined, then $dh=f'(g(x))g'(x)dx$.

(8) If $h(x)=e^{f(x)}$, then $dh=e^{f(x)}f'(x)dx$.

(9) If $f(x)>0$ for all x and $g(x)=\log(f(x))$, then $dg=\frac{f'(x)}{f(x)}dx$.

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